

A Level Mechanics

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Chapter 8: Vectors in Kinematics

This chapter is the capstone of the Mechanics unit. It takes the scalar suvat work of Chapter 2 (constant acceleration), the calculus methods of Chapter 4 (variable acceleration) and the projectile motion of Chapter 7, and rewrites all of it in $\mathbf{i}$, $\mathbf{j}$ vector notation. Nothing physically new is being introduced here — motion in two dimensions is still governed by the same definitions of displacement, velocity and acceleration you already know. What changes is the notation: instead of tracking horizontal and vertical motion with two separate sets of equations, you now track both at once with a single vector equation whose two components carry the same information.

Why this chapter matters

So far, every equation of motion in this course has described movement along a single straight line. Real motion — a thrown ball, a car turning a corner, a drone flying — happens in a plane, and describing it well requires a language that can hold “how far” and “in what direction” in one object. That language is vectors.

The good news is that almost nothing here is conceptually new. Every rule from Chapters 2, 4 and 7 still applies; it now simply applies to each of the two components (the $\mathbf{i}$ -part and the $\mathbf{j}$ -part) separately, then the results are combined back into a single vector. If you can already differentiate, integrate and apply suvat to one-dimensional motion, this chapter is mostly about learning a new notation for ideas you already understand — plus a few genuinely new skills, such as finding magnitude, direction and bearings from components, which this chapter treats in full.

8.1 Position, velocity and acceleration vectors

Concept

In one dimension a particle's position is a single signed number, s . In two dimensions its position is a **position vector**, $\mathbf{r}$, which gives both how far *and* in what direction the particle is from a fixed origin O — a single object that replaces “distance” and “direction” as two separate pieces of information.

Writing $\mathbf{i}$ and $\mathbf{j}$ for unit vectors along the x - and y -axes, a position vector is written in **$\mathbf{i}$, $\mathbf{j}$ notation**:

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j}$$

Here x and y are themselves functions of time, t — as the particle moves, both components change. The **velocity vector** $\mathbf{v}$ and **acceleration vector** $\mathbf{a}$ are defined exactly as in one dimension (velocity is the rate of change of position; acceleration is the rate of change of velocity), but now each component is differentiated separately:

$$\mathbf{v} = d\mathbf{r}/dt, \quad \mathbf{a} = d\mathbf{v}/dt$$

Every scalar suvat idea from Chapter 2 still applies — it now simply applies separately to the $\mathbf{i}$ -component and the $\mathbf{j}$ -component, and the two results are written as one vector at the end.

Two further quantities are calculated *from* a vector, rather than being part of how it is written:

- The **magnitude** of a vector is its size — a single non-negative scalar, found using Pythagoras' theorem on its components.
- The **direction** of a vector is the angle it makes with a fixed direction, usually the positive x -axis ($\mathbf{i}$), or expressed as a compass bearing.

A velocity vector's magnitude is exactly what Chapters 2 and 7 called “speed” — the two ideas are the same thing, just arrived at differently.

Key vocabulary

Term	Meaning
Position vector, $\mathbf{r}$	The displacement of a particle from a fixed origin O at time t , written $x\mathbf{i} + y\mathbf{j}$
Unit vectors $\mathbf{i}, \mathbf{j}$	Vectors of magnitude 1 along the positive x - and y -axes respectively
Component	The coefficient of $\mathbf{i}$ or $\mathbf{j}$ in a vector (the x -component or y -component)
Magnitude, $\mathbf{v}$	The size (a scalar, always ≥ 0) of a vector; for $\mathbf{v} = a\mathbf{i} + b\mathbf{j}$, $ \mathbf{v} = \sqrt{a^2 + b^2}$
Direction	The angle a vector makes with a fixed direction (often the positive x -axis, or a bearing)
Resultant vector	The single vector found by adding two or more vectors component-wise

Key formulas

KEY FORMULAS

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j}$$

$$|\mathbf{v}| = \sqrt{a^2 + b^2} \quad \text{for } \mathbf{v} = a\mathbf{i} + b\mathbf{j}$$

$$\theta = \arctan(b/a) \quad (\text{angle of } \mathbf{v} \text{ to the positive } x\text{-axis})$$

Worked example

EXAMPLE 8.1

Question. A particle has position vector $\mathbf{r} = (3t^2 - 2)\mathbf{i} + (5t)\mathbf{j}$ metres at time t seconds, $t \geq 0$. Find (a) the velocity vector $\mathbf{v}$ at time t , (b) the speed of the particle when $t = 2$, (c) the direction of motion when $t = 2$, as a bearing.

Step 1 (part a) — differentiate each component separately. Velocity is $d\mathbf{r}/dt$, so differentiate the $\mathbf{i}$ -component and the $\mathbf{j}$ -component of $\mathbf{r}$ one at a time:

- $\mathbf{i}$ -component: $d/dt(3t^2 - 2) = 6t$
- $\mathbf{j}$ -component: $d/dt(5t) = 5$

M1 A1

$$\mathbf{v} = 6t\mathbf{i} + 5\mathbf{j}$$

Step 2 (part b) — substitute, then find the magnitude. First substitute $t = 2$ into $\mathbf{v}$:

$$\mathbf{v} = 6(2)\mathbf{i} + 5\mathbf{j} = 12\mathbf{i} + 5\mathbf{j}$$

M1

Speed is the magnitude of the velocity vector, so apply Pythagoras' theorem to its two components:

$$|\mathbf{v}| = \sqrt{(12^2 + 5^2)} = \sqrt{(144 + 25)} = \sqrt{169} = 13 \text{ m s}^{-1}$$

A1

Step 3 (part c) — find the angle, then convert to a bearing. First find the angle θ the velocity vector makes with $\mathbf{i}$ (the positive x-axis), using the two components from Step 2:

$$\theta = \arctan(5/12) = 22.6^\circ \text{ to } \mathbf{i}$$

M1

A bearing is measured clockwise from north, not from the x-axis, so this angle must be converted. Here $\mathbf{i}$ points due east, so the bearing is found by subtracting the angle from 90° :

$$\text{Bearing} = 90^\circ - 22.6^\circ = 067.4^\circ$$

Rounding to three significant figures, the bearing is 067° . **A1**

Answer: (a) $\mathbf{v} = 6t\mathbf{i} + 5\mathbf{j}$. (b) 13 m s^{-1} . (c) 067° (3 s.f.).

Exam tip
EXAM TIP

It is common to arrive at this topic treating a 2D velocity as “a speed, plus a rough sense of direction” rather than as a genuine object with two independent numbers attached to it. Remember that $\mathbf{v} = 6t\mathbf{i} + 5\mathbf{j}$ is the full and complete answer to “find the velocity” — magnitude and direction are separate quantities calculated **from** the vector afterwards, not alternative ways of writing it. This distinction matters because a question that asks for velocity and receives only a speed, or vice versa, typically loses marks even when the underlying working is correct.

The bearing conversion in Step 3 is also worth flagging as a recurring source of error: it is easy to find an angle to a horizontal axis but forget the extra step of converting that angle to a compass bearing measured clockwise from north. Treat this conversion as a checklist item every time a bearing is requested, not something to remember only when it is prompted.

Common exam mistakes

- Quoting a velocity vector as a single number (its magnitude) instead of in $\mathbf{i}, \mathbf{j}$ form when the question asks for the velocity.
- Forgetting to take the square root when finding magnitude, or mixing up magnitude with one component.

- Muddling the angle to the x -axis with a compass bearing (forgetting to subtract from 90° , or not adjusting for the correct quadrant).

8.2 Using calculus with vectors

Concept

Because $\mathbf{i}$ and $\mathbf{j}$ are fixed, constant vectors, differentiating or integrating a vector function is done **component-wise**: differentiate (or integrate) the $\mathbf{i}$ -component and the $\mathbf{j}$ -component separately, exactly as in Chapter 4, then recombine the two results into a single vector. This means every variable-acceleration technique from Chapter 4 carries over unchanged, just applied twice — once per component:

- differentiating displacement to get velocity,
- differentiating velocity to get acceleration,
- integrating acceleration to get velocity (plus a constant vector, found from initial conditions),
- integrating velocity to get displacement (plus a constant vector, found from initial conditions).

When the acceleration is **constant**, the vector suvat equations mirror the scalar ones from Chapter 2 exactly, with every scalar quantity replaced by its vector equivalent:

$$\mathbf{v} = \mathbf{u} + \mathbf{a}t$$

$$\mathbf{r} = \mathbf{r}_0 + \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$$

Two further ideas appear often at A Level. The first is finding the time at which a particle is moving **parallel to a given direction**: this happens when the velocity's $\mathbf{i}$ - and $\mathbf{j}$ -components are in the same ratio as that direction's components. The second is finding the **speed** at a given instant, which is simply $|\mathbf{v}|$ at that time — the vector equivalent of the scalar “speed” already used throughout Chapters 2 and 7.

Key vocabulary

Term	Meaning
Component-wise differentiation/integration	Differentiating or integrating the $\mathbf{i}$ - and $\mathbf{j}$ -components of a vector separately
Constant vector of integration	The vector $\mathbf{c}$ added after integrating; found using given initial conditions (e.g. $\mathbf{r}$ at $t = 0$)
Initial position vector, $\mathbf{r}_0$	The position vector of the particle at $t = 0$
Parallel to a given vector	Velocity (or displacement) components in the same ratio as the components of the given direction vector
Speed	The magnitude $ \mathbf{v} $ of the velocity vector at a given instant (a scalar)

Key formulas

KEY FORMULAS

$$\mathbf{v}(t) = \int \mathbf{a}(t) dt + \mathbf{c}, \quad \mathbf{r}(t) = \int \mathbf{v}(t) dt + \mathbf{c}$$

$$\mathbf{r} = \mathbf{r}_0 + \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2 \quad (\text{constant acceleration only})$$

Worked example

EXAMPLE 8.2

Question. A particle starts at the point with position vector $2\mathbf{i} - \mathbf{j}$ metres and has constant acceleration $(3\mathbf{i} + 2\mathbf{j}) \text{ m s}^{-2}$. Its initial velocity is $(-\mathbf{i} + 4\mathbf{j}) \text{ m s}^{-1}$. Find (a) the position vector of the particle at time t , (b) the value of t at which the particle is moving parallel to $\mathbf{i}$.

Step 1 (part a) — identify each vector, then substitute. From the question:

$$\mathbf{r}_0 = 2\mathbf{i} - \mathbf{j}, \quad \mathbf{u} = -\mathbf{i} + 4\mathbf{j}, \quad \mathbf{a} = 3\mathbf{i} + 2\mathbf{j}$$

Substitute these into the vector suvat equation $\mathbf{r} = \mathbf{r}_0 + \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$, and simplify the $\mathbf{i}$ - and $\mathbf{j}$ -components separately. **M1**

$$\mathbf{i}\text{-component: } 2 + (-1)t + \frac{1}{2}(3)t^2 = 2 - t + 1.5t^2 \quad \mathbf{j}\text{-component: } -1 + (4)t + \frac{1}{2}(2)t^2 = -1 + 4t + t^2$$

$$\mathbf{r} = (2 - t + 1.5t^2)\mathbf{i} + (-1 + 4t + t^2)\mathbf{j}$$

A1

Step 2 (part b) — differentiate to get velocity, then set the correct component to zero.

Differentiate $\mathbf{r}$ term by term to find $\mathbf{v}(t)$:

$$\mathbf{v}(t) = (-1 + 3t)\mathbf{i} + (4 + 2t)\mathbf{j}$$

Moving parallel to $\mathbf{i}$ means the motion has no component in the $\mathbf{j}$ -direction at that instant, so the $\mathbf{j}$ -component of velocity must be zero. **M1**

$$4 + 2t = 0 \implies t = -2$$

This value is rejected, since the question requires $t \geq 0$. Checking the $\mathbf{j}$ -component of velocity directly confirms why: it equals 4 at $t = 0$ and keeps increasing for all $t \geq 0$, so it is never zero on the domain given — the particle is never moving parallel to $\mathbf{i}$ for $t \geq 0$. **A1**

Answer: (a) $\mathbf{r} = (2 - t + 1.5t^2)\mathbf{i} + (-1 + 4t + t^2)\mathbf{j}$. (b) No solution for $t \geq 0$ — the particle never travels parallel to $\mathbf{i}$ on this motion.

Exam tip

EXAM TIP

A common error at this stage is trying to “average” the $\mathbf{i}$ - and $\mathbf{j}$ -components together, or to differentiate the magnitude $|\mathbf{r}|$ directly, instead of differentiating each component and only combining them afterwards. Set out the working in two clearly separated lines — one for $\mathbf{i}$, one for $\mathbf{j}$ — until the component-wise habit becomes automatic, since this is the single most important mechanical skill in the whole chapter.

It is also worth deliberately including a “no solution” example, as above: A Level questions do not guarantee that “when is the particle moving in direction X” has a positive answer, and checking the working domain (here, $t \geq 0$) is an easy mark to drop if it is treated as an afterthought rather

than a required final step.

Common exam mistakes

- Integrating or differentiating the whole vector as if it were one scalar expression, instead of component-wise.
- Omitting the constant vector of integration, or finding only one of its two components from the initial conditions.
- Setting the wrong component to zero when finding when motion is parallel to **i** or **j** (parallel to **i** means the **j**-component is zero, and vice versa).

8.3 Projectiles using vectors

Concept

Chapter 7 modelled projectile motion using separate horizontal and vertical scalar suvat equations. In vector form, exactly the same physics — constant velocity horizontally, constant acceleration g downwards vertically — is written as a **single vector equation for the position**, $\mathbf{r}(t)$, with the acceleration vector $\mathbf{a} = -g\mathbf{j}$ (taking **j** as vertically upward):

$$\mathbf{r}(t) = \mathbf{r}_0 + \mathbf{u}t + \frac{1}{2}(-g\mathbf{j})t^2$$

If a particle is projected from the origin with speed U at angle θ above the horizontal, its initial velocity vector is $\mathbf{u} = U \cos \theta \mathbf{i} + U \sin \theta \mathbf{j}$, and the position vector becomes:

$$\mathbf{r}(t) = (U \cos \theta)t\mathbf{i} + [(U \sin \theta)t - \frac{1}{2}gt^2]\mathbf{j}$$

This is not new content — it is the Chapter 7 horizontal equation $x = (U \cos \theta)t$ sitting inside the **i**-component, and the Chapter 7 vertical equation $y = (U \sin \theta)t - \frac{1}{2}gt^2$ sitting inside the **j**-component. The advantage of the vector form is that it lets you differentiate or integrate the whole projectile motion at once (as in Section 8.2), and it generalises immediately to projectiles launched with *any* initial velocity vector, not just ones described as “speed and angle”.

Key vocabulary

Term	Meaning
Position vector of a projectile, $\mathbf{r}(t)$	$x(t)\mathbf{i} + y(t)\mathbf{j}$, combining the horizontal and vertical scalar equations of Chapter 7 into one vector
Acceleration due to gravity (vector form)	$\mathbf{a} = -g\mathbf{j}$ (taking j vertically upward; g is the usual positive constant)
Initial velocity vector, $\mathbf{u}$	$U \cos \theta \mathbf{i} + U \sin \theta \mathbf{j}$, where U is the launch speed and θ the launch angle
Time of flight	The value(s) of t for which the j -component of $\mathbf{r}$ equals the landing height
Velocity vector at time t, $\mathbf{v}(t)$	The derivative of $\mathbf{r}(t)$; its magnitude gives the projectile's speed at time t

Key formulas

KEY FORMULAS

$$\mathbf{r}(t) = (U \cos \theta)t\mathbf{i} + [(U \sin \theta)t - \frac{1}{2}gt^2]\mathbf{j}$$

$$\mathbf{v}(t) = U \cos \theta\mathbf{i} + (U \sin \theta - gt)\mathbf{j}$$

Worked example

EXAMPLE 8.3

Question. A particle is projected from a point O on horizontal ground with velocity $(8\mathbf{i} + 6\mathbf{j}) \text{ m s}^{-1}$, where $\mathbf{i}$ and $\mathbf{j}$ are horizontal and vertical unit vectors respectively. Using $g = 9.8 \text{ m s}^{-2}$, find (a) the position vector of the particle at time t , in terms of t , (b) the time at which the particle returns to the height of O, (c) the speed of the particle at that time.

Step 1 (part a) — substitute into the constant-acceleration vector suvat equation. Here $\mathbf{u} = 8\mathbf{i} + 6\mathbf{j}$, $\mathbf{a} = -9.8\mathbf{j}$, and $\mathbf{r}_0 = \mathbf{0}$ (the particle starts at O). Substitute into $\mathbf{r} = \mathbf{ut} + \frac{1}{2}\mathbf{at}^2$, and simplify each component. **M1**

i-component: $8t$ (no acceleration horizontally) j-component: $6t + \frac{1}{2}(-9.8)t^2 = 6t - 4.9t^2$

$$\mathbf{r}(t) = 8t\mathbf{i} + (6t - 4.9t^2)\mathbf{j}$$

A1

Step 2 (part b) — set the j-component to zero, then solve. The particle returns to the height of O exactly when its vertical displacement is zero, i.e. when the j-component of $\mathbf{r}$ is zero. **M1**

$$6t - 4.9t^2 = 0$$

Factorise:

$$t(6 - 4.9t) = 0$$

This gives $t = 0$ (the point of projection, rejected as trivial) or:

$$t = 6/4.9 = 1.2245\dots$$

A1

Step 3 (part c) — differentiate to get velocity, substitute, then find the magnitude.

Differentiate $\mathbf{r}(t)$ to get $\mathbf{v}(t) = 8\mathbf{i} + (6 - 9.8t)\mathbf{j}$, then substitute $t = 1.2245$: **M1**

$$\mathbf{v} = 8\mathbf{i} + (6 - 9.8 \times 1.2245)\mathbf{j} = 8\mathbf{i} - 6\mathbf{j}$$

Apply Pythagoras' theorem to find the speed:

$$|\mathbf{v}| = \sqrt{(8^2 + 6^2)} = \sqrt{(64 + 36)} = \sqrt{100} = 10 \text{ m s}^{-1}$$

A1

Answer: (a) $\mathbf{r}(t) = 8t\mathbf{i} + (6t - 4.9t^2)\mathbf{j}$. (b) $t = 6/4.9 = 1.22 \text{ s}$ (3 s.f.). (c) 10 m s^{-1} .

Exam tip

EXAM TIP

By this point you should recognise this as the same type of projectile met in Chapter 7, with $U \cos \theta = 8$ and $U \sin \theta = 6$, now solved with vectors instead of separate horizontal/vertical

working. It is worth deliberately re-solving one earlier scalar projectile problem in vector form as a bridging exercise, so you see the two methods side by side rather than as unrelated topics.

It is also a useful, exam-relevant observation that the landing speed here (10 m s^{-1}) equals the launch speed $|\mathbf{u}| = \sqrt{8^2 + 6^2} = 10 \text{ m s}^{-1}$. This is not a coincidence: on level ground, the launch and landing speeds are always equal, because gravity does zero net work over a journey that returns to the same height. This symmetry, met informally in Chapter 7, is something vector methods make straightforward to prove in general, either using energy or by observing that the $\mathbf{j}$ -component of velocity is simply reversed in sign between launch and landing.

Common exam mistakes

- Writing $\mathbf{a} = 9.8\mathbf{j}$ instead of $-9.8\mathbf{j}$ (a sign error from not fixing $\mathbf{j}$ as “upward” consistently through the whole solution).
- Solving $|\mathbf{r}| = 0$ for “return to the height of O” instead of setting only the $\mathbf{j}$ -component to zero.
- Reporting the final velocity as a vector when the question asks for speed, or vice versa — always check whether magnitude or the full vector is required.

How this chapter ties the unit together

This chapter is deliberately a synthesis, not new physics. Section 8.1 rewrites the constant- and variable-motion definitions of **Chapter 2** and **Chapter 4** in two dimensions. Section 8.2 shows that the calculus techniques of **Chapter 4** — differentiating and integrating to move between displacement, velocity and acceleration — apply unchanged, component-wise, to vectors. Section 8.3 shows that the projectile equations of **Chapter 7** are simply the $\mathbf{i}$ - and $\mathbf{j}$ -components of a single vector position function.

If you are confident with Chapters 2, 4 and 7 in scalar form, and have internalised this chapter's component-wise habit, you have everything needed to answer any A Level (or Further Maths) two-dimensional kinematics question — including ones that mix projectiles with variable acceleration, or ask for a resultant force or displacement expressed as a vector.

Chapter summary

Idea	Key point
Position, velocity, acceleration vectors	$\mathbf{r} = x\mathbf{i} + y\mathbf{j}$, $\mathbf{v} = d\mathbf{r}/dt$, $\mathbf{a} = d\mathbf{v}/dt$ — generalises Chapters 2 and 4 to two dimensions
Magnitude and direction	$ \mathbf{v} = \sqrt{a^2 + b^2}$ gives speed; direction found from arctan of the components, converted to a bearing if required
Vector calculus	Differentiate/integrate $\mathbf{i}$ - and $\mathbf{j}$ -components separately; constant vector of integration found from initial conditions
Constant acceleration (vector form)	$\mathbf{r} = \mathbf{r}_0 + \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$ and $\mathbf{v} = \mathbf{u} + \mathbf{a}t$, the vector versions of Chapter 2's suvat equations

Projectiles in vector form	$\mathbf{r}(t) = (U \cos \theta)t \mathbf{i} + [(U \sin \theta)t - \frac{1}{2}gt^2] \mathbf{j}$, unifying Chapter 7's horizontal and vertical equations
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Common exam mistakes

- Treating a vector equation as if it were scalar — adding or equating magnitudes instead of working component-wise.
- Forgetting to take the square root (or the correct root) when converting from components back to magnitude/speed.
- Arithmetic slips when combining **i**- and **j**-components separately, especially sign errors with a downward acceleration written as $-g\mathbf{j}$.
- Giving a vector where a magnitude (speed, distance) was asked for, or a magnitude where a full vector (velocity, position) was asked for.

You should be able to...

- Write position, velocity and acceleration as vectors in **i, j** notation, and find one from another by differentiation or integration.
- Find the magnitude and direction (including as a bearing) of a vector given in **i, j** form.
- Apply the vector forms of the constant-acceleration equations, $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ and $\mathbf{r} = \mathbf{r}_0 + \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$, including finding when a particle moves parallel to a given direction.
- Express projectile motion as a single vector position function $\mathbf{r}(t)$, linking it explicitly to the separate horizontal/vertical scalar equations of Chapter 7.
- Recognise that the component-wise habit built in this chapter — treating **i**- and **j**-parts separately and combining only at the end — is the single skill that carries every other technique in this chapter.